

Le Fang

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RESEARCH INTERESTS

My research sits at the intersection of **human-computer interaction**, **ubiquitous computing** and **psychology**, and asks how large language models can **make humans more human**: helping them understand themselves, care for their wellbeing and connect with one another. I study what LLMs can infer about people from their **everyday digital traces** (smartphone sensor data, including screen text), where those inferences go wrong, and what changes when models enter the communication between people.

EDUCATION

The University of Melbourne — PhD in Human-Computer Interaction Oct 2023 – present

- School of Computing and Information Systems; thesis submission expected April 2027
- Advisors: Prof. Vassilis Kostakos, Prof. Jorge Goncalves, Dr. Hong Jia

Northwestern University — M.S. in Computer Science Sep 2020 – Dec 2022

- GPA 3.83 / 4.00

Pennsylvania State University — B.S. in Information Sciences & Technology Aug 2014 – Dec 2018

- GPA 3.72 / 4.00; Cum Laude; Schreyer Honors Scholar

RESEARCH EXPERIENCE

Visiting Researcher, The University of Tokyo — IIS Lab, host: Prof. Koji Yatani Jun 2026 – Jul 2026

- Member of ASPIRE-MWI (Mental Well-being Intelligence), a JST ASPIRE-funded international research community on data-driven mental well-being, collaborating on LLMs and mental health.
- Led a scoping review of LLM-mediated communication in human-human relationships (364 peer-reviewed papers), mapping which relationships LLMs are entering and with what consequences, using an AI-first review process with researcher verification.

PhD Research: Personalising LLMs with Passive Sensing — The University of Melbourne Oct 2023 – present

- Personalised an LLM with 17 weeks of passive smartphone sensing data from 83 participants and evaluated its inferences of affect against weekly self-reports, using history-based baselines, ordinal metrics and cluster-bootstrap confidence intervals; identified a systematic positivity bias.
- Ran a longitudinal ablation study of smartphone sensor modalities to identify which signals predict personality traits.
- Evaluated how adding digital phenotyping data to LLM prompts changes the quality of AI-generated life advice, combining human and automated evaluation.
- Designed an agentic framework that ingests on-screen text and other sensor data, reasons over user context and goals, and generates actionable recommendations.

AWARE Digital Phenotyping Platform — The University of Melbourne Sep 2023 – present

- Contribute features and performance improvements to the open-source mobile sensing platform; built the analysis pipeline used across multiple longitudinal studies.
- Built Aware Narrator, which converts raw sensor data into human-readable narrative diaries (github.com/awareframework/aware_narrator).

PUBLICATIONS

1. Fang, L., Zhang, T., Brockmeier, L. C., Zhang, S., Fei, R., Ma, Y., Teng, S., D'Alfonso, S., Koval, P., Jia, H., Goncalves, J., & Kostakos, V. (in press). Large language models rate human happiness higher than humans do. *npj Digital Public Health*.
2. Ma, Y., Govers, J., Fang, L., Zhang, S., Hu, Y. O., Yi, X., & Goncalves, J. (2026). "Black Mirror?": Public sensemaking of AI-powered lifelogging wearables. In *Companion of the 2026 ACM International Joint Conference on Pervasive and Ubiquitous Computing and the 2026 ACM International Symposium on Wearable Computers (UbiComp/ISWC '26)*. ACM.

3. **Fang, L.**, Zhang, S., Jia, H., Goncalves, J., & Kostakos, V. (2024). ScreenTK: Seamless detection of time-killing moments using continuous mobile screen text and on-device LLMs. In *Companion of the 2024 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp '24)*. ACM. <https://doi.org/10.1145/3675094.3677547>
4. Zhang, T., Zhang, S., **Fang, L.**, Jia, H., Kostakos, V., & D'Alfonso, S. (2024). AutoJournaling: A context-aware journaling system leveraging MLLMs on smartphone screenshots. In *Proceedings of the 30th Annual International Conference on Mobile Computing and Networking (MobiCom '24)*, EIFCom Workshop (pp. 2347–2352). ACM. <https://doi.org/10.1145/3636534.3698122> **Best Presentation Award.**
5. Zhang, S., Ma, Y., **Fang, L.**, Jia, H., D'Alfonso, S., & Kostakos, V. (2024). Enabling on-device LLMs personalization with smartphone sensing. In *Companion of the 2024 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp '24)*. ACM. <https://doi.org/10.1145/3675094.3677545>
6. Calvillo, J., **Fang, L.**, Cole, J., & Reitter, D. (2020). Surprisal predicts code-switching in Chinese-English bilingual text. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Association for Computational Linguistics. <https://doi.org/10.18653/v1/2020.emnlp-main.330>

TEACHING AND SUPERVISION

- **Head Tutor**, Mobile Computing, The University of Melbourne 2026
- **Casual Tutor**, The University of Melbourne — Mobile Computing, Media Computation, UI Development (FEIT); Statistical Learning (Melbourne Business School) 2024 – present
- **Research Project Supervisor**, Master of Computing and Information Systems project "Bilingual Social Media User Behavior Patterns" 2024

ACADEMIC SERVICE

- **Social Chair**, EIFCOM 2026: International Workshop on Mobile Systems with Efficient Foundation Models 2026
- **Reviewer**, *PLOS Digital Health*; *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* (IMWUT)
- **Session Host**, 3-Minute Paper session, 9th CIS Doctoral Colloquium, The University of Melbourne 2023
- **Volunteer**, University of Melbourne Open Day 2024

SCHOLARSHIPS AND HONORS

- Melbourne Research Scholarship, The University of Melbourne 2023 – present
- Schreyer Honors Scholar, Pennsylvania State University 2014 – 2018

SKILLS

- **Methods**: LLM-based system design, agentic workflows, multimodal LLMs, on-device deployment of quantized LLMs, mobile passive sensing, longitudinal field studies, experimental design and statistical modeling
- **Programming**: Python (PyTorch, Hugging Face Transformers, NumPy, Pandas), Java / Android, R, JavaScript, SQL
- **Languages**: English (proficient), Mandarin (native)